

Makine Öğrenmesi- Uygulama

Ders 12



Bu Dersin İçeriği

I. Göğüs Kanseri Teşhisi Karar Ağacı Uygulaması

Bu Dersin İçeriği

- ❑ Bu meme kanseri hastaları veri seti, nüfus temelli kanser istatistikleri hakkında bilgi sağlayan NCI'nin SEER (Gözetim, Epidemiyoloji ve Sonuçlar) Programının 2017 Kasım güncellemesinden elde edilmiştir.
- ❑ Veri seti, 2006-2010 yıllarında infiltratif duktal ve lobüler karsinom meme kanseri (SEER birincil siteler NOS histoloji kodları 8522/3) tanısı konulan kadın hastaları içermektedir.
- ❑ Tümör boyutu bilinmeyen, bölgesel LN'leri incelenen, pozitif bölgesel LN'leri olan ve hayatta kalma süresi 1 aydan az olan hastalar hariç tutulmuştur; böylece, nihai olarak 4024 hasta dahil edilmiştir.

Uygulama

File - Orange

Source

File: Ders12/Breast_Cancer.xlsx

Sheet: Breast_Cancer

File Type: Automatically detect type

Info

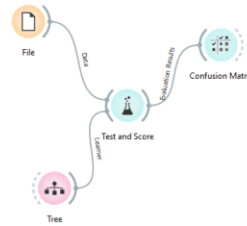
4024 instances
16 features (0.0% missing values)
Data has no target variable.
0 meta attributes

Columns (Double click to edit)

Name	Type	Role	Values
1 Age	numeric	feature	
2 Race	categorical	feature	Black, Other, White
3 Marital Status	categorical	feature	Divorced, Married, Separated, Single, Widowed
4 T Stage	categorical	feature	T1, T2, T3, T4
5 N Stage	categorical	feature	N1, N2, N3
6 M Stage	categorical	feature	IA, IB, IIA, IIB, IIIC
7 differentiate	categorical	feature	Moderately differentiated, Poorly differentiated, Undifferentiated, Well differentiated
8 Grade	numeric	feature	
9 A Stage	categorical	feature	Distant, Regional
10 Tumor Size	numeric	feature	
11 Estrogen Status	categorical	feature	Negative, Positive
12 Progesterone Status	categorical	feature	Negative, Positive
13 Regional Node Examined	numeric	feature	
14 Regional Node Positive	numeric	feature	
15 Survival Months	numeric	feature	
16 Status	categorical	target	Alive, Dead

Reset Apply

Browse documentation datasets



Confusion Matrix - Orange

Learned

Tree

Output

☒ Predictions
☐ Probabilities
☒ Apply Automatically

Select Correct Select Misclassified Clear Selection

		Predicted		Σ
		Alive	Dead	
Actual	Alive	3303	129	3432
	Dead	299	294	593
Σ		3602	423	4025

Tree - Orange

Name: Tree

Parameters

☒ Induce binary tree

☒ Min. number of instances in leaves: 10

☒ Do not split subsets smaller than: 5

☒ Limit the maximal tree depth to: 100

Classification

☒ Stop when majority reaches [%]: 95

☒ Apply Automatically

Test and Score - Orange

File Edit View Window Help

☐ Cross validation
Number of folds: 10
☐ Stratified

☐ Cross validation by feature

☒ Random sampling
Repeat train/test: 5
Training set size: 80 %
☐ Stratified

☐ Leave one out
☐ Test on train data
☐ Test on test data

Evaluation results for target: (None, show average over classes)

Model	AUC	CA	F1	Prec	Recall	MCC	Spec	LogLoss
Tree	0.788	0.894	0.886	0.884	0.894	0.530	0.565	0.694

Compare models by: Area under ROC curve

☐ Negligible diff.: 0.1

Tree

Side shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Bu Dersin İçeriği

- ❑ Veri seti kaynağı:
- ❑ <https://www.kaggle.com/datasets/reihanenamdari/breast-cancer?resource=download>
- ❑ <https://seer.cancer.gov/about/>